

AI-Collaboration Log: Project 2

Khoi Nguyen

Overview

In this course, we will treat AI as a collaborator, not a substitute for thought. This document serves as a log for your collaboration with AI tools (e.g., ChatGPT, Gemini, Claude, etc.) during Project 2. Please be specific in your descriptions. **Failure to document AI usage or representing AI-generated work as entirely your own will be considered an honor code violation.**

Mention the AI tool(s) you used:

ChatGPT Claude

1. Idea Generation

Describe how you used AI to brainstorm your topic, find potential datasets, or refine your research questions. What were some typical prompts that you used? How did the AI's output influence your project? Me and Tiana decided to work with the Formula 1 World Championship dataset from Kaggle because I was interested in F1 racing, and the dataset had a rich variety of related CSV files that formed a natural relational structure. We explored the three tables to understand what variables were available, then each brainstormed questions we were curious about. Together we came up with around 10 questions, used ChatGPT to evaluate whether each was meaningful and feasible, then refined them down to 8 final questions so each of us will do 4.

Throughout the project we used Claude and ChatGP to help us understand SQL concepts, structure our queries, and format our code.

2. Code Generation (SQL & R)

Describe how AI assisted in writing, debugging, or optimizing your code. How did you verify and correct AI-generated code? Did the AI provide any incorrect or “hallucinated” code? What did you think were some benefits of AI-assisted coding for this project? What were some challenges?

I have asked Claude and ChatGPT questions about the logic behind what we were trying to do mostly. We used it to understand the structure first, and then we drafted the code ourselves in a separate file, tested it, and only added it to our branch after it worked.

AI was helpful when we were debugging our code. Instead of spending a long time searching through every line whenever we have errors, we paste the error message into ChatGPT, and it helped us find the issue. After that, we still fixed the code ourselves and checked that it ran correctly before committing it.

The biggest challenge was dealing with merge conflicts. At the beginning, we kept running into conflicts when pushing and pulling through Posit Cloud and Git, and we were not sure how to resolve them. We showed ChatGPT the conflict errors and used its steps as a guide to fix them

3. Final Reflection

In separate paragraphs, reflect on (1) how the use of AI changed your learning process for this project, and (2) your overall experience with this project (challenges/achievements/skills you learned, etc.).

AI helped me understand the dataset better, explore possible ideas, recognize errors faster, and get advice on how to structure my code more clearly. Since I am new to SQL, it was useful to have support when I was unsure how to organize a query or think through the logic behind it. I did not think this project was super challenging, but it was a good starting point because it gave me practice with SQL in a real dataset and helped me feel more comfortable testing ideas, finding mistakes, and learning how the code works.